Normative ethics for AI alignment

A critical annotated bibliography for *The Frontier of Normative Alignment: Advancements in Ethics-Driven Large Language Models (2025-2026)*.

27 cited items

Research papers, preprints, repositories, practitioner material, and discovery sources assessed for contribution, use, and limitations.

Prepared 11 October 2026

Based on the user-supplied Gemini Deep Research report.

READER'S GUIDE

The evidence is broad, but not evenly strong

The report assembles a timely research map: moral-rationale evaluation, framework-specific reinforcement learning, pluralistic preference aggregation, Rawlsian fairness, legal alignment, multi-agent constitutions, activation steering, and bidirectional human-AI adaptation. Its strongest claims point to identifiable research papers. Its weakest chain is the Rawlsian section, where community posts and repository pages appear beside primary research.

21

research-facing papers, preprints, reviews, posters, or submissions

6

secondary, repository, practitioner, duplicate, or index entries needing cautious use

1

clear duplicate: items 16 and 17 refer to the same arXiv record

5

practical layers: evaluate, train, aggregate, govern, monitor

> **How to use this bibliography:** “Primary research” describes source type, not final certainty. Most 2025-2026 arXiv items are recent preprints. Confirm the current version, venue, methods, and corrections before relying on a result in a publication, policy, or production safety case.

Annotation lens

- **Contribution:** what the source adds to the report's argument.
- **Best use:** the decision, evaluation, or design question it can inform.
- **Limitation:** what must be checked before carrying the claim forward.

Sources by report theme

1. Evaluations of AI Moral Reasoning Still Miss Half of the Picture
2. Divergent Moral Grounds in Human and LLM Ethical Judgments
3. Generalizable Moral Decision Alignment For LLM Agents Using ...
4. Moral Alignment for LLM Agents
5. Where Norms and References Collide: Evaluating LLMs on ...
6. PlurVA-LLM-2026 Shared Task Track-1
7. Operationalizing Pluralistic Values in Large Language Model ...
8. AI Alignment From Social Choice Perspectives
9. A Roadmap to Impactful Pluralistic Alignment Research
10. Adaptive Pluralistic Alignment: A pipeline for dynamic ...
11. Using John Rawls' veil of ignorance to align AI with fairness principles
12. Rawlsian Agents: An Application of Large Language Models (LLM) to Forge Fairer Bilateral Agreements
13. The veil of ignorance idea proposes that if we can't know ...
14. Using the Veil of Ignorance to align AI systems with principles of justice
15. Legal Alignment for Safe and Ethical AI
16. Semantic Alignment Between Normative Theories of Ethics ...
17. Semantic Alignment Between Normative Theories of Ethics and the ...
18. Diverse Human Value Alignment for Large Language Models ...
19. Statutory AI: Aligning Large Language Models With Legal Norms
20. Evolving Interpretable Constitutions for Multi-Agent Coordination
21. Constitutional Midtraining: Content Presence Drives Alignment Gains
22. Activation Steering in 2026: A Practitioner's Field Guide
23. Activation Steering via Contrastive Causal Mediation
24. Revealing the Intrinsic Ethical Vulnerability of Aligned Large ...
25. Reverse Constitutional AI: A Framework for Controllable Toxic Data
26. Position: Towards Bidirectional Human-AI Alignment
27. NeurIPS 2025 Papers

1. Evaluations of AI Moral Reasoning Still Miss Half of the Picture

PRIMARY RESEARCH / ARXIV PREPRINT

Frames the distinction between matching moral labels and aligning the reasons behind them. It supplies the report's central diagnostic: verdict agreement can coexist with a different moral vocabulary and therefore weak out-of-distribution reliability. Use it to justify rationale-level evaluation, not merely accuracy scoring. As a recent preprint, its benchmark design, annotation protocol, and later revisions should be checked before treating its numerical results as settled.

https://arxiv.org/pdf/2608.14566

2. Divergent Moral Grounds in Human and LLM Ethical Judgments

PRIMARY RESEARCH / ARXIV ABSTRACT RECORD

Provides the empirical comparison of human and model rationales across commonsense morality, deontology, justice, utilitarianism, and virtue ethics. The report uses it for domain-specific differences such as models' over-weighting of generic harm or respectfulness. This is directly relevant to explanation quality and safety-case evidence. Confirm the full paper's sample composition and coding reliability before generalizing beyond the reported Central European annotator pool.

https://arxiv.org/abs/2608.12368

3. Generalizable Moral Decision Alignment For LLM Agents Using ...

PRIMARY RESEARCH / ARXIV ABSTRACT RECORD

Introduces the MoralReason approach: supervised format training followed by reasoning-level reinforcement learning for explicit normative frameworks. It underpins the report's claim that utilitarian and deontological reasoning can generalize to disjoint scenarios when both decisions and reasoning traces are rewarded. The approach is technically actionable, but framework-conformant traces are not the same as philosophical correctness; inspect the reward model and adversarial tests.

https://arxiv.org/abs/2511.12271

4. Moral Alignment for LLM Agents

PRIMARY RESEARCH / ARXIV HTML

Studies moral alignment in interactive agents, including environments where choices are represented through payoff structures. The report uses it as supporting evidence that intrinsic rewards can sustain distinct normative strategies over repeated interactions. Its value is in connecting moral theory to agent behavior rather than single-turn text. Transfer from simplified games to open social environments remains uncertain, so treat it as mechanism evidence rather than deployment validation.

https://arxiv.org/html/2410.01639v1

5. Where Norms and References Collide: Evaluating LLMs on ...

PRIMARY RESEARCH / ARXIV HTML

Presents the SNIC diagnostic setting for resolving ambiguous references through prescriptive and prohibitive social norms in embodied tasks. It anchors the report's warning that abstract ethical fluency does not guarantee grounded norm use in nearly identical physical scenes. This source is especially useful for robotics and human-agent interaction evaluations. The benchmark is diagnostic and narrow; performance should not be interpreted as a complete measure of embodied social competence.

https://arxiv.org/html/2602.02975v1

6. PlurVA-LLM-2026 Shared Task Track-1

SHARED-TASK PAPER / ARXIV PREPRINT

Documents a multilingual pluralistic-value task spanning Chinese, Indonesian, and Sinhala. The report draws on it for threshold calibration and targeted augmentation in lower-resource settings, including the stated macro-average improvement for Sinhala. It is a useful benchmark-design case study. Shared-task results can be system- and split-specific, so compare task definitions, baselines, and evaluation scripts before claiming broad cross-cultural alignment gains.

https://arxiv.org/pdf/2609.32382

7. Operationalizing Pluralistic Values in Large Language Model ...

PRIMARY RESEARCH / ARXIV PREPRINT

Examines how diverse human ratings can be retained rather than collapsed into a single consensus target. It supports the report's demographic-disagreement discussion and the argument for richer rating scales. The contribution is practical: preference collection and aggregation choices become alignment design variables. Because demographic categories and rating dimensions can be context-sensitive, review sampling, measurement invariance, and subgroup uncertainty before using the reported effect sizes in policy.

https://arxiv.org/pdf/2511.14476

8. AI Alignment From Social Choice Perspectives

CONCEPTUAL AND TECHNICAL REVIEW / ARXIV PREPRINT

Recasts feedback aggregation as a social-choice problem, exposing limitations of majority voting and pairwise preference models. It helps explain why apparently neutral reinforcement learning from human feedback can erase minorities or create unstable collective preferences. This is a strong conceptual bridge between political theory and alignment engineering. Its prescriptions still require empirical choices about whose preferences count, which voting rule is legitimate, and how strategic reporting is handled.

https://arxiv.org/pdf/2606.21550

9. A Roadmap to Impactful Pluralistic Alignment Research

RESEARCH AGENDA / ARXIV PREPRINT

Organizes open problems in pluralistic alignment and clarifies the difference between population-level pluralism and personalization. The report relies on this distinction to avoid treating one user's customization as representation of a diverse public. It is most useful for scoping programs, metrics, and stakeholder processes. As a roadmap, it synthesizes priorities rather than proving that any specific aggregation or deployment method works.

https://arxiv.org/pdf/2607.22305

10. Adaptive Pluralistic Alignment: A pipeline for dynamic ...

SYSTEMS PROPOSAL / ARXIV PREPRINT

Proposes a modular pipeline that updates aggregation as values change, aiming to prevent value lock-in while reusing expensive model-training components. The report cites it as an architecture for dynamic social choice. It is valuable for lifecycle governance because it treats alignment as maintenance, not a one-time target. Evaluate update stability, auditability, gaming resistance, and the institutional authority behind preference refreshes before production use.

https://arxiv.org/pdf/2605.01642

11. Using John Rawls' veil of ignorance to align AI with fairness principles

COMMUNITY POST / SECONDARY POINTER

Offers an accessible discussion of applying Rawls's veil of ignorance to AI fairness. It may help locate the underlying study or explain the intuition to a general audience, but it is not an adequate scholarly source for empirical claims. Do not cite it for sample sizes, effect estimates, or experimental methods. Replace it with the original paper or an official research page wherever the argument is load-bearing.

https://www.facebook.com/groups/8208901822/posts/10161024869541823/

12. Rawlsian Agents: An Application of Large Language Models (LLM) to Forge Fairer Bilateral Agreements

REPOSITORY COPY / RESEARCH MANUSCRIPT

Explores LLM agents constrained by a Rawlsian state of ignorance when negotiating bilateral agreements. It supports the report's claim that computational agents can repeatedly reason under explicit informational constraints. The manuscript is relevant to contract design and bargaining simulations. ResearchGate is a hosting page, not a quality signal; verify authorship, version, publication venue, experimental setup, and any conflicts between simulated fairness and enforceable legal fairness.

https://www.researchgate.net/publication/398661671_Rawlsian_Agents_An_Application_of_Large_Language_Models_LLM_to_Forge_Fairer_Bilateral_Agreements

13. The veil of ignorance idea proposes that if we can't know ...

COMMUNITY DISCUSSION / SECONDARY POINTER

Summarizes public discussion around a veil-of-ignorance experiment and can help identify questions raised by non-specialists. It should not be used as evidence for the report's human-participant findings because comments and post summaries are not controlled scholarship. Use only as a discovery aid, then cite the original study. The presence of a science-community label does not validate the accuracy of the linked discussion.

https://www.reddit.com/r/science/comments/13nvtq2/the_veil_of_ignorance_idea_proposes_that_if_we/

14. Using the Veil of Ignorance to align AI systems with principles of justice

REPOSITORY COPY / RESEARCH MANUSCRIPT

Reports human-AI experiments in which participants select governance principles without knowing their future position, supporting priority for the worst-off. It anchors the report's treatment of Rawls's Difference Principle as an elicitation device for AI objectives. The study is central to the fairness section. Verify the version of record, preregistration, treatment wording, participant recruitment, and whether outcomes persist outside the experimental framing.

https://www.researchgate.net/publication/370229416_Using_the_Veil_of_Ignorance_to_align_AI_systems_with_principles_of_justice

15. Legal Alignment for Safe and Ethical AI

LEGAL SCHOLARSHIP / ARXIV PREPRINT

Defines legal alignment as designing AI systems to operate according to legal rules, interpretive methods, and institutional concepts. The report uses its three-part taxonomy to position law as a democratically grounded, externally inspectable lower bound. It is a strong organizing source for governance architecture. Law is jurisdiction-specific, contested, and incomplete; legal compliance cannot substitute for moral legitimacy or resolve conflicts between statutes and rights.

https://arxiv.org/html/2601.04175v1

16. Semantic Alignment Between Normative Theories of Ethics ...

PRIMARY COMPUTATIONAL ANALYSIS / ARXIV PDF

Uses semantic textual similarity to compare the European Union AI Act with utilitarian, deontological, and virtue-ethical texts. It supports the report's claim that the Act's language aligns most closely with deontological concepts. The source is useful for hypothesis generation about regulatory fit. Semantic proximity is not doctrinal interpretation: embedding choices, corpus construction, and section weighting can drive results and should not be read as legal intent.

https://arxiv.org/pdf/2601.13372

17. Semantic Alignment Between Normative Theories of Ethics and the ...

DUPLICATE RECORD / ARXIV ABSTRACT

This entry appears to be the abstract-page version of item 16 rather than an independent source. It offers convenient metadata and version history but no additional evidence. Consolidate items 16 and 17 in formal citation lists, retaining one canonical record and noting the consulted version. Counting both would overstate the size of the evidence base.

https://arxiv.org/abs/2601.13372

18. Diverse Human Value Alignment for Large Language Models ...

PRIMARY RESEARCH / ARXIV PREPRINT

Investigates alignment across heterogeneous human judgments and supports the report's quantitative discussion of demographic differences in toxicity, emotional awareness, sensitivity, and helpfulness ratings. It makes visible how aggregate objectives can encode the dominant rater population. The source is directly relevant to dataset governance. Interpret subgroup findings with attention to sample size, category construction, multiplicity, and the risk of essentializing demographic groups.

https://arxiv.org/pdf/2511.00379

19. Statutory AI: Aligning Large Language Models With Legal Norms

PRIMARY SYSTEMS RESEARCH / ARXIV PREPRINT

Tests alignment against explicit statutory themes rather than broad constitutional principles. The report cites substantial reductions in harmful content and lower computation relative to a Constitutional AI baseline. This is promising evidence that operational legal categories can improve classification and refusal behavior. Before adoption, audit jurisdictional coverage, legal updates, false positives, adversarial robustness, and whether benchmark reductions translate into legally correct decisions.

https://arxiv.org/pdf/2608.28593

20. Evolving Interpretable Constitutions for Multi-Agent Coordination

PRIMARY MULTI-AGENT RESEARCH / ARXIV HTML

Treats a constitution as an evolvable symbolic object in a resource-sharing simulation. It supports the report's findings that concrete operational rules can outperform abstract values and sharply reduce explicit coordination while improving system-level outcomes. The work is valuable because its learned rules remain inspectable. Grid-world gains may not survive richer incentives, deception, communication channels, or institutional power asymmetries; external validation is essential.

https://arxiv.org/html/2602.00755v1

21. Constitutional Midtraining: Content Presence Drives Alignment Gains

PRIMARY TRAINING RESEARCH / ARXIV PREPRINT

Studies inserting constitutional material during late pretraining rather than relying only on post-hoc safety tuning. The report uses it to argue that mid-training can create more durable, projectable dispositions that survive later benign fine-tuning. It is relevant to training-stack design and persistence tests. Examine capability trade-offs, data contamination, alignment-faking evaluations, and whether the effects generalize across model families and constitutions.

https://arxiv.org/pdf/2607.26654

22. Activation Steering in 2026: A Practitioner's Field Guide

PRACTITIONER SYNTHESIS / TECHNICAL BLOG

Explains contrastive activation vectors, layer selection, strength control, and common deployment patterns. The report uses it to distinguish steerable low-dimensional behaviors such as refusal or tone from complex properties such as truthfulness and moral reasoning. It is useful as an implementation overview. Because it is a practitioner guide rather than peer-reviewed evidence, verify claims against the cited experiments and reproduce results on the target model.

https://subhadipmitra.com/blog/2026/activation-steering-field-guide/

23. Activation Steering via Contrastive Causal Mediation

RESEARCH SUBMISSION / OPENREVIEW

Provides a causal-mediation approach to identifying and applying steering directions inside model activations. It grounds the report's mathematical description of a contrastive steering vector and the limits of single-direction interventions. The source is technically relevant for inference-time control. Review the paper's current decision, revisions, reviewer concerns, layer-ablation evidence, and failure cases; OpenReview records can change during the review cycle.

https://openreview.net/forum?id=n25T76QbxQ

24. Revealing the Intrinsic Ethical Vulnerability of Aligned Large ...

PRIMARY ADVERSARIAL EVALUATION / ARXIV HTML

Argues that post-hoc alignment suppresses harmful behavior without removing underlying harmful representations. The report cites its adversarial results to characterize safety tuning as a fragile behavioral wrapper. This source is important for threat modeling and distribution-shift testing. Scrutinize the attack model, target set, judge calibration, and definition of attack success before carrying the headline rate into claims about all aligned systems.

https://arxiv.org/html/2504.05050

25. Reverse Constitutional AI: A Framework for Controllable Toxic Data

PRIMARY METHODS RESEARCH / ARXIV PREPRINT

Develops critique-and-revision loops for synthesizing controlled toxic or adversarial training data under a constitution. The report presents it as one response to intrinsic vulnerability and reward hacking. Its practical value lies in producing targeted data while preserving a readable normative scaffold. The method itself creates dual-use material; governance should cover access, leakage, evaluator independence, and whether synthetic attacks represent real adversaries.

https://arxiv.org/pdf/2604.17769

26. Position: Towards Bidirectional Human-AI Alignment

CONFERENCE POSTER / POSITION PAPER

Argues that alignment must consider how humans adapt to AI, not only how AI adapts to human preferences. It supports the report's closing shift toward reciprocal learning, trust, reliance, and long-term co-adaptation. This expands evaluation from model outputs to sociotechnical trajectories. As a position paper, it identifies a research gap rather than establishing a complete causal model; pair it with longitudinal human-subjects evidence.

https://neurips.cc/virtual/2025/poster/121940

27. NeurIPS 2025 Papers

CONFERENCE INDEX / DISCOVERY SOURCE

Provides the conference-wide paper index and context for the bidirectional-alignment poster. It is useful for locating related peer-reviewed work but is not itself evidence for any substantive claim. Cite the specific paper or poster instead of the index. In a cleaned bibliography, retain this only as a discovery aid or remove it to avoid inflating the source count.

https://neurips.cc/virtual/2025/papers.html

What the bibliography supports

Measure reasons, not only decisions

Items 1-5 establish the report's core diagnostic: a model can choose a human-preferred answer while relying on a materially different rationale. Safety evaluations should therefore score action choice, stated reasons, counterfactual consistency, and transfer to unseen contexts.

Train explicit frameworks, then test their projectability

Items 3-5 and 20-21 support structured reasoning and constitutional interventions, but they also show that benchmark success can be local. The central engineering question is whether a learned disposition remains coherent under new tasks, incentives, fine-tuning, and multi-agent interaction.

Preserve disagreement rather than hiding it

Items 6-10 and 18 frame alignment as collective choice. Rating scale, aggregation rule, language, demographic composition, and update mechanism are part of the normative system. No technical optimizer removes the need for legitimate governance.

Use law as a floor, not the whole moral target

Items 15-19 show why statutory categories and legal interpretation are attractive: they are explicit and contestable. Yet law varies by jurisdiction, can lag moral change, and can encode injustice. Legal alignment should complement rather than replace ethical analysis and rights review.

Keep inference-time controls in their lane

Items 22-25 support activation steering and adversarial data generation as useful control tools. They do not establish deep moral competence or reliable truthfulness. Complex normative reasoning still requires sequential computation, robust evaluation, and institutional oversight.

Turn the reading list into an evaluation program

1. **De-duplicate and version-lock.** Merge items 16 and 17; record authors, revision date, venue, and persistent identifier for every paper.
2. **Replace discovery sources.** Use items 11, 13, and 27 only to find the underlying paper. Locate versions of record for items 12 and 14.
3. **Reproduce headline numbers.** Recheck dataset size, splits, effect definitions, uncertainty, and evaluator setup for every percentage quoted in the report.
4. **Build a rationale-level test suite.** Include moral verdict, principle selection, explanation consistency, counterfactual invariance, and out-of-distribution cases.
5. **Stratify human feedback.** Preserve individual ratings and report subgroup uncertainty rather than using only majority labels.
6. **Test across languages and settings.** Separate translation quality, threshold calibration, cultural variation, and low-resource data effects.
7. **Stress legal and constitutional layers.** Evaluate conflicting rules, jurisdiction changes, prompt injection, authority abuse, and multi-agent incentives.
8. **Monitor co-adaptation.** Measure how repeated interaction changes human trust, reliance, decision quality, and willingness to challenge the system.

> **Bottom line:** The bibliography supports a layered safety architecture, not a single “ethical model” score. Stronger use of this evidence requires primary-source cleanup, version control, and direct replication of the report's quantitative claims.